

Implementation Plan – AI-Based Wellness Recommendation System

1. Overview

This application is a wellness platform that provides therapy recommendations using an AI-powered chatbot. The system is built using Spring Boot with Spring AI, where the AI logic is directly integrated into the backend.

The chatbot interacts with the user by asking structured questions to collect symptom details and then generates personalized recommendations.

2. Architecture

The system follows a simple architecture:

- Frontend: React.js (Chatbot UI)
- Backend: Spring Boot (Business logic + AI integration)
- AI Layer: Spring AI with Gemini 2.5 flash LLM
- Database: MySQL

3. Modules

3.1 Chatbot Module

- Collects user health inputs step-by-step:
 - Symptoms
 - Severity level (low/medium/high)
 - Duration (how long)
 - Additional symptoms

3.2 AI Recommendation Module

- Uses Spring AI + Gemini
- Generates therapy suggestions based on collected inputs

4. Chatbot Flow (Beginner Logic)

When the user opens the chatbot:

1. Bot asks:
 - What symptoms do you have?

- What is the severity level?
 - How long have you had this issue?
 - Any additional symptoms?
- 2. User answers step-by-step
- 3. Backend collects all responses
- 4. All data is combined and sent to AI
- 5. AI generates final recommendation

5. API Endpoints

AI Recommendation

- POST /ai/recommend
→ Accepts user symptom details and returns AI response

Chatbot

- POST /chat/start
- POST /chat/answer

Notifications

- GET /notifications/{userId}

6. AI Pipeline (Spring AI + Gemini)

Flow:

1. User input is collected
2. Backend prepares structured data
3. External API data is fetched
4. Data is passed to Gemini using Spring AI
5. Gemini generates response
6. Response is returned to user

7. Prompt Template

You are a health assistant.

User Details:

- Symptoms: {symptoms}
- Severity Level: {level}
- Duration: {duration}
- Additional Symptoms: {extra}

Instructions:

- Understand the user's condition based on the given details.
- Replicate OpenFDA, WHO guidelines, and fitness applications for better response to the user.
- Provide safe and simple therapy recommendations.
- Include basic health advice and lifestyle suggestions.
- Avoid complex medical terms.
- If the condition seems serious, suggest consulting a doctor.

Generate a clear, helpful, and user-friendly response.

9. Technologies Used

- Backend: Spring Boot
- AI Integration: Spring AI + Gemini
- Frontend: React.js
- Database: MySQL

10. Conclusion

This system provides a simple and effective way to generate AI-based wellness recommendations using a chatbot interface. By integrating AI directly into the backend and using structured user inputs, the application remains easy to implement while still delivering intelligent and personalized results.